

CENTRE FOR ADVANCED STUDIES

**DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH,
LUCKNOW**



EVALUATION SCHEME AND SYLLABI

For

B.Tech. Ist Year
Computer Science & Engineering
Computer Science & Engineering (AI/ML)

Effective from the Session: 2025-26

DR. A.P.J ABDUL KALAM TECHNICAL UNIVERSITY UTTAR PRADESH.
Sec-11, Jankipuram Vistar, Lucknow-226031, Uttar Pradesh, India

DETAILED SYLLABI

B.Tech. First Year

Effective from Session 2025-26

B.Tech. (Computer Science & Engineering) First Year, Semester-I

3-WEEKS STUDENT INDUCTION PROGRAMME

In the beginning of the session

										Evaluation Scheme			
SN	Subject Code	Subject Name	Type	Category	Period			Sessional Component		Sessional (SW)(TS/PS)	End Semester Examination (ESE)	Total	Credit
					L	T	P	CT	TA	CT+TA	TE/PE	SW+ESE	Cr
1.	CAS 103	Engineering Mathematics-I	T	BS	3	0	0	20	10	30	70	100	3
2.	CCS 101	Introduction to Computer Science	T	ES	3	0	0	20	10	30	70	100	3
3.	CEE 102	Electrical & Electronics Engineering	T	ES	3	1	0	20	10	30	70	100	4
4.	CCS 102	Computational Thinking through Programming	T	ES	3	1	0	20	10	30	70	100	4
5.	CAS 104	Environment & Energy	T	BS	3	0	0	20	10	30	70	100	3
6.	CME 152	Engineering Visualization & Design Lab	P	ES	0	0	3	-	50	50	50	100	1
7.	CEE 152	Electrical & Electronics Engineering Lab	P	ES	0	0	2	-	50	50	50	100	1
8.	CCS 152	Computational Thinking through Programming Lab	P	ES	0	0	2	-	50	50	50	100	1
9.	CME 151	IDEA Lab Workshop	P	ES	0	0	6	-	50	50	50	100	2
					15	2	13			350	550	900	22
10.	Vocational Training / Skill based courses (Mandatory for Exit after I year)				-	-	-	-	-	-	-	-	2

Abbreviation Used:

BS: Basic Science Course

ES: Engineering Science Course

HS: Humanities and Social Science Course

VA: Value Added Course

B.Tech. (Computer Science & Engineering) First Year, Semester-II

								Evaluation Scheme					
SN	Subject Code	Subject Name	Type	Category	Period			Sessional Component		Sessional (SW)(TS/PS)	End Semester Examination (ESE)	Total	Credit
					L	T	P	CT	TA	CT+TA	TE/PE	SW+ESE	Cr
1.	CAS 203	Engineering Mathematics-II	T	BS	3	0	0	20	10	30	70	100	3
2.	CCS 201	Computer Organization	T	ES	3	1	0	20	10	30	70	100	4
3.	CCS 204	Web Programming and Python	T	ES	3	1	0	20	10	30	70	100	4
4.	CCS 203	Data Structures	T	ES	3	1	0	20	10	30	70	100	4
5.	CAS 205	English for Technical Writing	T	HS	3	0	0	20	10	30	70	100	3
6.	CCS 251	Computer Organization Lab	P	ES	0	0	2	-	50	50	50	100	1
7.	CCS 254	Web Programming and Python Lab	P	ES	0	0	2	-	50	50	50	100	1
8.	CCS 253	Data Structure Lab	P	ES	0	0	2	-	50	50	50	100	1
9.	CAS 253	English Language Lab	P	HS	0	0	2	-	50	50	50	100	1
10.	CVA 252	Sports and Yoga/NSS	P	VA	0	0	3*	-	100	*100	-	*100	0
					15	3	08+3*			350+100*	550	900+100*	22
11.	Vocational Training / Skill based courses (Mandatory for Exit after I year)				-	-	-	-	-	-	-	-	2

*Compulsory Qualifying Audit Course

Abbreviation Used:

BS: Basic Science Course

ES: Engineering Science Course

HS: Humanities and Social Science Course

VA: Value Added Course

Summer Internship (4-week)/NPTEL Course(4 week) during summer break after Semester-II and same will be assessed/evaluated in the Semester-III

B.Tech. Computer Science & Engineering (AI/ML) First Year, Semester-I

3-WEEKS STUDENT INDUCTION PROGRAMME

In the beginning of the session

								Evaluation Scheme					
SN	Subject Code	Subject Name	Type	Category	Period			Sessional Component		Sessional (SW)(TS/PS)	End Semester Examination (ESE)	Total	Credit
					L	T	P	CT	TA	CT+TA	TE/PE	SW+ESE	Cr
1.	CAS 103	Engineering Mathematics-I	T	BS	3	0	0	20	10	30	70	100	3
2.	CAI 101	Introduction to Computer Science and Artificial Intelligence	T	ES	3	0	0	20	10	30	70	100	3
3.	CEE 102	Electrical & Electronics Engineering	T	ES	3	1	0	20	10	30	70	100	4
4.	CCS 102	Computational Thinking through Programming	T	ES	3	1	0	20	10	30	70	100	4
5.	CAS 104	Environment & Energy	T	BS	3	0	0	20	10	30	70	100	3
6.	CME 152	Engineering Visualization & Design Lab	P	ES	0	0	3	-	50	50	50	100	1
7.	CEE 152	Electrical & Electronics Engineering Lab	P	ES	0	0	2	-	50	50	50	100	1
8.	CCS 152	Computational Thinking through Programming Lab	P	ES	0	0	2	-	50	50	50	100	1
9.	CME 151	IDEA Lab Workshop	P	ES	0	0	6	-	50	50	50	100	2
					15	2	13			350	550	900	22
10.	Vocational Training / Skill based courses (Mandatory for Exit after 1 year)				-	-	-	-	-	-	-	-	2

Abbreviation Used:

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B.Tech. Computer Science & Engineering (AI/ML) First Year, Semester-II

								Evaluation Scheme					
SN	Subject Code	Subject Name	Type	Category	Period			Sessional Component		Sessional (SW)(TS/PS)	End Semester Examination (ESE)	Total	Credit
					L	T	P	CT	TA	CT+TA	TE/PE	SW+ESE	Cr
1.	CAS 203	Engineering Mathematics-II	T	BS	3	0	0	20	10	30	70	100	3
2.	CCS 201	Computer Organization	T	ES	3	1	0	20	10	30	70	100	4
3.	CCS 204	Web Programming and Python	T	ES	3	1	0	20	10	30	70	100	4
4.	CCS 203	Data Structures	T	ES	3	1	0	20	10	30	70	100	4
5.	CAS 205	English for Technical Writing	T	HS	3	0	0	20	10	30	70	100	3
6.	CCS 251	Computer Organization Lab	P	ES	0	0	2	-	50	50	50	100	1
7.	CCS 254	Web Programming and Python Lab	P	ES	0	0	2	-	50	50	50	100	1
8.	CCS 253	Data Structure Lab	P	ES	0	0	2	-	50	50	50	100	1
9.	CAS 253	English Language Lab	P	HS	0	0	2	-	50	50	50	100	1
10.	CVA 252	Sports and Yoga/NSS	P	VA	0	0	3*		100	*100	-	*100	0
					15	3	08+3*			350+100*	550	900+100*	22
11.	Vocational Training / Skill based courses (Mandatory for Exit after I year)				-	-	-	-	-	-	-	-	2

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CAS 103: ENGINEERING MATHEMATICS-I

Course Objectives:

The objective of this course is to familiarize the graduate engineers with techniques in matrix, calculus, multivariate analysis and vector calculus. It aims to equip the students with standard concepts and tools from intermediate to advanced level that will enable them to tackle more advanced level of mathematics and applications that they would find useful in their disciplines.

The students will learn:

- The essential tools of matrices, Eigen values and its application in a Comprehensive manner.
- To apply the knowledge of differential calculus in the field of engineering.
- To deal with functions of several variables that is essential in optimizing the results of real-life problems.
- To apply integral calculus in various field of engineering and have a basic understanding of Beta and Gamma functions and application of Dirichlet's integral.
- To deal with vector calculus that is required in different branches of Engineering to graduate engineer.

Content	Contact Hours
Unit-1: Matrices	8
Elementary transformations, Inverse of a matrix, Rank of matrix, Solution of system of linear equations, Characteristic equation, Cayley-Hamilton Theorem and its application, Linear Dependence and Independence of vectors, Eigen values and Eigen vectors, Complex Matrices, Hermitian, Skew-Hermitian and Unitary Matrices, Applications to Engineering problems.	
Unit-2: Differential Calculus- I	8
Successive Differentiation (nth order derivatives), Leibnitz theorem, Curve tracing, Partial derivatives, Euler's Theorem for homogeneous functions, Total derivative, Change of variables.	
Unit-3: Differential Calculus-II	8
Expansion of functions by Taylor's and Maclaurin's theorems for functions of one and two variables, Maxima and Minima of functions of several variables, Lagrange's method of multipliers, Jacobians, Approximation of errors.	
Unit-4: Multiple Integration	8
Double integral, Triple integral, Change of order of integration, Change of variables, Beta and Gamma function and their properties, Dirichlet's integral and its applications to area and volume, Liouville's extensions of Dirichlet's integral.	
Unit-5: Vector Calculus	8

Vector Differentiation: Gradient, Curl and Divergence and their Physical interpretation, Directional derivatives. Vector Integration: Line integral, Surface integral, Volume integral, Gauss's Divergence theorem, Green's theorem and Stoke's theorem (without proof) and their applications.	
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Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Understand the concept of complex matrices, Eigen values, Eigen vectors and apply the concept of rank to evaluate linear simultaneous equations	K2, K5
CO 2	Remember the concept of differentiation to find successive differentiation, Leibnitz Theorem, and create curve tracing, and find partial and total derivatives	K1, K6 & K5
CO 3	Applying the concept of partial differentiation to evaluate extrema, series expansion, error approximation of functions and Jacobians	K3, K5
CO 4	Remember the concept of Beta and Gamma function; analyze area and volume and Dirichlet's theorem in multiple integral	K1, K4
CO 5	Apply the concept of Vector Calculus to analyze and evaluate directional derivative, line, surface and volume integrals.	K3, K4 & K5

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books:

1. B. V. Ramana, Higher Engineering Mathematics, McGraw-Hill Publishing Company Ltd., 2008.
2. B. S. Grewal, Higher Engineering Mathematics, Khanna Publisher, 2005.
3. R K. Jain & S R K. Iyenger, Advance Engineering Mathematics, Narosa Publishing House 2002.

Reference Books:

1. E. Kreyszig, Advance Engineering Mathematics, John Wiley & Sons, 2005.
2. Peter V. O'Neil, Advance Engineering Mathematics, Thomson (Cengage) Learning, 2007.
3. Maurice D. Weir, Joel Hass, Frank R. Giordano, Thomas, Calculus, Eleventh Edition, Pearson.
4. D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2005.
5. Veerarajan T., Engineering Mathematics for first year, McGraw-Hill, New Delhi, 2008.
6. Ray Wylie C and Louis C Barret, Advanced Engineering Mathematics, McGraw-Hill; Sixth Edition.
7. P. Sivaramakrishna Das and C. Vijayakumari, Engineering Mathematics, 1st Edition, Pearson Education.
8. Advanced Engineering Mathematics. Chandrika Prasad, Reena Garg, 2018.
9. H.K. Dass, Advanced Engineering Mathematics, H.K.Dass, S. Chand

CAS 203: ENGINEERING MATHEMATICS-II

Course Objectives:

The objective of this course is to familiarize the prospective engineers with techniques in ordinary differential equations, Laplace transform, sequence and series, Fourier series and complex variables. It aims to equip the students to deal with advanced level of mathematics and applications that would be essential for their disciplines.

The students will learn:

- The effective mathematical tools for the solutions of differential equations that model physical processes.
- The basic knowledge of Laplace transforms and its applications in solving differential equations.
- The tool for convergence of series and expansion of function using Fourier series for learning advanced Engineering Mathematics.
- The tools of differentiation of functions of complex variables that are used in various techniques dealing with engineering problems.
- The tools of integration of functions of complex variables that are used in various techniques dealing with engineering problems.

Content	Contact Hours
Unit-1: Ordinary Differential Equation of Higher Order	8
Linear differential equation of n th order with constant coefficients, Simultaneous linear differential equations, Second order linear differential equations with variable coefficients, Solution by changing independent variable, Method of variation of parameters, Cauchy-Euler equation, Application of differential equations in solving engineering problems.	
Unit-2: Laplace Transform	8
Laplace transform, Existence theorem, Properties of Laplace Transform, Laplace transform of derivatives and integrals, Unit step function, Laplace transform of periodic function, Inverse Laplace transform, Convolution theorem. Application of Laplace Transform to solve ordinary differential equations and simultaneous differential equations.	
Unit-3: Sequence and Series	8
Definition of Sequence and series with examples, Convergence of series, Tests for convergence of series, Ratio test, D' Alembert's test, Raabe's test, Comparison test. Fourier series, Half range Fourier sine and cosine series.	
Unit-4: Complex Variable–Differentiation	8
Functions of complex variable, Limit, Continuity and differentiability, Analytic functions, Cauchy- Riemann equations (Cartesian and Polar form), Harmonic function, Method to find Analytic functions, Milne's Thompson Method, Conformal mapping, Mobius transformation and their properties.	
Unit-5: Complex Variable –Integration	8
Complex integration, Cauchy- Integral theorem, Cauchy integral formula, Taylor's and Laurent's series, singularities and its classification, zeros of analytic functions, Residues, Cauchy's Residue theorem and its application.	

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Remember the concept differentiation to evaluate LDE of nth order with constant coefficient and LDE with variable coefficient of 2nd order.	K1 & K5
CO 2	Understand and apply the concept of Laplace Transform to evaluate differential equations.	K2 ,K3 & K5
CO 3	Understand the concept of convergence to analyze the convergence of series and expansion of the function for Fourier series.	K2 & K4
CO 4	Apply the concept of analyticity, Harmonic function and create the image of function applying conformal transformation.	K3, K6& K3
CO 5	Apply the concept of Cauchy Integral theorem, Cauchy Integral formula, singularity and calculus of residue to evaluate integrals.	K3& K5

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books:

1. B. V. Ramana, Higher Engineering Mathematics, Tata McGraw-Hill Publishing Company Ltd., 2008.
2. B. S. Grewal, Higher Engineering Mathematics, Khanna Publisher, 2005.
3. 3. R. K. Jain & S. R. K. Iyenger, Advance Engineering Mathematics ,Narosa Publishing - House, 2002.

Reference Books:

1. E. Kreyszig, Advance Engineering Mathematics, John Wiley & Sons, 2005.
2. Peter V. O'Neil, Advance Engineering Mathematics, Thomson (Cengage) Learning, 2007.
3. Maurice D. Weir, Joel Hass, Frank R. Giordano, Thomas, Calculus, Eleventh Edition, Pearson.
4. G.B Thomas, R L Finney, Calculus and Analytical Geometry, Ninth Edition Pearson, 2002.
5. James Ward Brown and Ruel V Churchill, Fourier Series and Boundary Value Problems, 8th Edition-McGraw-Hill
6. D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2005.
7. Veerarajan T., Engineering Mathematics for first year, McGraw-Hill, New Delhi, 2008.
8. Charles E Roberts Jr, Ordinary Differential Equations, Application, Model and Computing, CRC Press T&F Group.
9. Ray Wylie C and Louis C Barret, Advanced Engineering Mathematics, 6th Edition, McGraw-Hill.
10. James Ward Brown and Ruel V Churchill, Complex Variable and Applications, 8th Edition, McGraw-Hill.

11. P. Sivaramakrishna Das and C. Vijayakumari, Engineering Mathematics, 1st Edition, Pearson India Education Services Pvt. Ltd.
12. Advanced Engineering Mathematics By Chandrika Prasad, Reena Garg Khanna Publishing House, Delhi.
13. Laplace Transforms by Schaum's series, 2005 Edition, Spiegel Publication.
14. H.K. Dass, Advanced Engineering Mathematics, H.K.Dass, S. Chand

CEE 102: ELECTRICAL AND ELECTRONICS ENGINEERING

Course Objectives:

The students will learn the basic concepts of electrical and electronics engineering. They will study about various types of electronics devices and electrical machines along with their fundamental aspects. This will be useful to the students who wish to pursue higher studies in electrical or electronics or relevant fields.

Content	Contact Hours
Unit-1	8
PN Diode and Transistor: PN Diode Theory, PN Characteristic and Breakdown Region, PN Diode Application as Rectifier. Zener Diode Theory, Zener Voltage Regulator, Diode as Clamper and Clipper, Photodiode Theory, LED Theoryand applications. Bipolar Junction Transistor Theory,Common Collector, Emitter and Base Configurations, Different Biasing Techniques, Concept of Transistor Amplifier, Introduction to FET Transistor and its feature.	
Unit-2	8
Rectifiers: Half wave, center tapped, bridge and full-wave rectifiers Transistors: Construction and characteristics of BJT, MOSFET Transistor configuration: CB, CE, CC configuration, Transistor at low frequency, small signal low frequency transistor model (h-parameters), Analysis of transistor amplifier using h-parameters, Transistor biasing and bias stabilization, Emitter resistance bias circuit and self-bias circuit, Bias compensation techniques.	
Unit-3	8
Voltage and current sources, dependent and independent sources, source conversion, DC circuit's analysis using mesh & nodal method, Thevenin's, and superposition theorem. 1-phase AC circuits under sinusoidal steady-state, active, reactive, and apparent power, physical meaning of reactive power, power factor.	
Unit-4	8
Transformers: Magnetic Circuits: Review of laws of electromagnetism, Flux, MMF and their relation, analysis of the magnetic and electric circuit. Singlephase transformer: Basic concepts, constructional features, EMF equation, voltage, current, and impedance transformation, Equivalent circuits.	
Unit-5	8
DC Machines: DC Machines: Constructional features, working principle and types of dc machinesand their characteristics. Induction Machines: Constructional features, working principle and types of induction motors. Synchronous Machines: Constructional features, working principle and emf equation	

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Understand basics of p-n junction diodes	K2
CO 2	To study the design and operation of rectifiers and transistor amplifiers.	K3
CO 3	To study and apply circuit theorems to AC and DC circuits.	K3
CO 4	Apply the concept of analyticity, Harmonic function and create the image of function applying conformal transformation.	K4
CO5	To understand the working principle of different machines	K2

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

TextBooks and Reference Books:

1. Ritu Sahdev, Basic Electrical Engineering, Khanna Book Publishing, 2022
2. I. J. Nagrath, D.P Kothari, Basic Electrical Engineering, Mc-Graw Hill Education, 2005.
3. A.Husain, Electric Machines, Dhanpat Rai & Co.
4. Hayt and Kimberly, Engineering Circuit Analysis, Tata Mc-Graw Hill Publishing, 2013
5. <https://onlinecourses.nptel.ac.in/>

CAS 104: ENVIRONMENT AND ENERGY

Course Objectives:

1. Aims and objectives of environmental education emphasize the relationship between man and the environment and educate young people about the importance of nature and the environment.
2. Environmental education aims to impart ecological knowledge and promote environmentally conscious behavior towards nature.
3. It encourages young minds to take responsibility for protecting the natural environment protection through information and knowledge and to develop environmental awareness.
4. Incidentally, promoting awareness and a sense of respect for nature leads to a comprehensive understanding of the environment and a reasonable attitude towards protecting it.
5. The focus of environmental education is Awareness, Knowledge, Attitude, Skills, Capacity Building and Participation.

Syllabus

Content	Contact Hours
UNIT I	8
Environment: Definition, Types of Environment, Components of environment, Segments of environment, Scope and importance, Need for Public Awareness. Ecosystem: Definition, Types of ecosystem, Structure of ecosystem, Food Chain, Food Web, Ecological pyramid. Balance Ecosystem. Effects of Human Activities such as Food, Shelter, Housing, Agriculture, Industry, Mining, Transportation, Economic and Social security on Environment, Environmental Impact Assessment, Sustainable Development.	
UNIT II	8
Pollution and their Effects; Public Health Aspects of Environmental; Water Pollution, Air Pollution, Soil Pollution, Noise Pollution, Thermal Pollution, Radioactive Pollution, Marine Pollution and Solid waste management.	
UNIT III	8
Current Environmental Issues of Importance; Global Warming, Green House Effects, Climate Change, Acid Rain, Ozone Layer Formation and Depletion, Population Growth and Automobile pollution, Burning of paddy straw, concept of carbon credit, CO ₂ sequestration, environmental impact assessment	
UNIT IV	8
Basics of Energy: Classification of energy sources, Global and national energy scenario, Fossil and alternate fuels and its characterization. Energy Conservation Systems: General aspects of energy conservation and management. Energy conversion systems: Working principle, Basic components, General functioning and normal rating specifications of various energy conversion systems like Power plant, Pump, Refrigerator, Air-conditioner, Internal combustion engine, Solar PV cell, Solar water heating system, Biogas plant. Wind turbine, Fuel cells.	
UNIT V	8
Environmental Protection: Environmental Protection Act 1986, Initiatives by	

Non Governmental Organizations (NGO's), Human Population and the Environment: Population growth, Environmental Education, Women Education.	
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Course Outcomes:

Upon completion of the course, the student will be able to:

	Course Outcomes	Bloom's Level
CO-1	Gain in-depth knowledge on natural processes that sustain life, and govern economy.	K2
CO-2	Differentiate between various types of environmental pollution along with their impacts and regulatory standards	K3
CO-3	Examine various global environmental issues and their management	K4
CO-4	Discuss the fundamental principles of energy, including classification, conservation and related policy frameworks and regulations. Get acquainted with the concept of energy systems and their components	K3
CO-5	Adopt sustainability as a practice in life, society and industry.	K3

TEXT BOOKS AND REFERENCE BOOKS:

1. Textbook of Environment and Ecology by Dave, Katewa & Singh, 2nd Edition, Cengage Learning India Pvt Ltd Delhi.
2. Daniel B. Botkin & Edward A Keller, Environmental Sciences, John Wiley & Sons.
3. R. Rajagopalan, Environmental Studies, Oxford University Press
4. Environmental Studies by S Deswal, Dhanpat Rai & Co.
5. <https://onlinecourses.nptel.ac.in/>

CCS 102: COMPUTATIONAL THINKING THROUGH PROGRAMMING

The objectives of this course are:

The objective of this course is to grow the computational thinking and problem-solving ability of students. Moreover, the aim of this subject is to create various programming concepts such as inputs/outputs, variables, control statements, functions, arrays, pointers, structures, etc. For coding or writing the programs, syntaxes of C language will be taught.

Syllabus

Content	Contact Hours
Unit-1: Digital computer and Programming	8
Introduction to Digital Computer and Programming (2 hours): Basic components of computer, binary representation, bits and bytes, program, software. Introduction to Computational Thinking (2 hours): Procedural computational approach to real life problems, idea of algorithms, creating flowcharts and pseudo-code. Introduction to Computational Problem Solving through C (2 hours): Programming language concepts and its applicability on problem solving, introducing C programming language, inputs and outputs, compiling and running C program	
Unit-2: Fundamentals of C	8
C Fundamentals (4 hours): C character set, identifiers and keywords, data type, consonants, declarations, operators (arithmetic, relational, logical, assignment, unary, bitwise, etc.). Control Statements (4 hours): Branching: if-else, Looping: while, do-while, for, nested control, switch, break, continue, goto.	
Unit-3: Functions and Variables	8
Functions (4 hours): Defining a function, accessing a function, function prototypes, argument passing, and recursion. Variables (3 hours): variable and their scopes, automatic, external/global, static variables.	
Unit-4: Arrays and Pointers	8
Arrays (4 hours): Defining an array, processing arrays, passing arrays to functions, multi-dimensional arrays. Pointers (4 hours): defining pointers, passing pointers to functions, dynamic memory allocation, operations on pointers, arrays of pointers call by value vs. call by reference.	
Unit-5: Structure and File Handling	8
Structure and Unions: Defining and processing a structure, user defined data types, structures and pointers, passing structure to functions, self-referential structures, Unions. Data Files: File handling, multi-file programming. String: Defining and processing string. Various operations on string.	

Program analysis: Debugging programs with gdb, memory analysis using valgrind.	
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Course Outcome:

At the end of course, the student will be able to

Course Outcome(CO)		Bloom's Level
CO1	Understand fundamentals of programming	K2
CO2	Understand and apply fundamentals of C language	K2, K3
CO3	Understand and implement functions and variables	K2, K3
CO4	Understand and apply the use of arrays and pointers	K2, K3
CO5	Understand and implement data files, string and program analysis	K2, K3

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create

Text Books and Reference Books:

1. Schaum's Outline of Programming with C by Byron Gottfried, McGraw-Hill
2. The C programming by Kernighan Brain W. and Ritchie Dennis M., Pearson Education.
3. Computer Basics and C Programming by V. Rajaraman, PHI Learning Pvt. Limited, 2015.
4. Computer Concepts and Programming in C, E Balaguruswami, McGraw-Hill
5. Computer Science-A Structured Programming Approach Using C, by Behrouz A. Forouzan, Richard F. Gilberg, Thomson, Third Edition, Cengage Learning - 2007.
6. Let Us C By Yashwant P. Kanetkar.
7. Problem Solving and Program Designing C, by Jeri R. Hanly, Elliot B. Koffman, Pearson Addison- Wesley, 2006.
8. Programming in C by Kochan Stephen G. Pearson Education-2015.
9. Computer Concepts and Programming in C by D.S.Yadav and Rajeev Khanna, New Age International Publication.
10. Computer Concepts and Programming by Anami, Angadi and Manvi, PHI Publication
11. Computer Concepts and Programming in C by Vikas Gupta, Wiley India Publication
12. Computer Fundamentals and Programming in C. Reema Thareja, Oxford Publication
13. <https://onlinecourses.nptel.ac.in/>

CCS 201: COMPUTER ORGANIZATION

Course Objectives: The objective of this course is to familiarize the students with electronic circuits and logic. Transistors, operational amplifiers, etc will be introduced along with Boolean algebra and logic circuits design. The core purpose of the course is to prepare the students for taking up advanced courses in similar domain.

Syllabus

Content	Contact Hours
Unit-1: Waveshaping Circuits And Operational Amplifier: Linear Wave Shaping Circuits, RC High Pass and Low Pass Circuits, RC Integrator and Differentiator Circuits, Nonlinear Wave Shaping Circuits, Two Level Diode Clipper Circuits, Clamping Circuits, Operational Amplifier OP-AMP with Block Diagram, Schematic Symbol of OP AMP, 741 Package Style and Pinouts, Specifications of Op-Amp, Inverting and Non-Inverting Amplifier, Voltage Follower Circuit, Multistage OP-AMP Circuit, OP-AMP Averaging Amplifier, OP-AMP Subtractor.	8
Unit-2: Boolean Algebra And Switching Functions: Basic Logic Operation and Logic Gates, Truth Table, Basic Postulates and Fundamental Theorems of Boolean Algebra, Standard Representations of Logic Functions- SOP and POS Forms, Simplification of Switching Functions-K-Map and Quine-Mccluskey Tabular Methods, Synthesis of Combinational Logic Circuits.	8
Unit-3: Introduction To Sequential Logic Circuits: Basic Concepts of Sequential Circuits; Cross Coupled SR Flip-Flop Using NAND or NOR Gates; JK Flip-Flop Rise Condition; Clocked Flip-Flop; D-Type and Toggle Flip-Flops; Truth Tables and Excitation Tables for Flip-Flops; Master Slave Configuration; Edge Triggered and Level Triggered Flip-Flops; Elimination of Switch Bounce using Flip-Flops; Flip-Flops with Preset and Clear.	8
Unit-4: Sequential Logic Circuit Design: Basic Concepts of Counters and Registers; Binary Counters; BCD Counters; Up Down Counter; Johnson Counter, Module-N Counter; Design of Counter Using State Diagrams and Table; Sequence Generators; Shift Left and Right Register; Registers with Parallel Load; Serial-In Parallel-Out (SIPO) And Parallel-In-Serial-Out (PISO); Register using Different Type of Flip-Flop.	8
Unit-5: Register Transfer Logic: Arithmetic, Logic and Shift Micro-Operation; Conditional Control Statements; Fixed-Point and Floating-Point Data; Arithmetic Shifts; Instruction Code and Design Of Simple Computer. Processor Logic Design: Processor Organization; Design of Arithmetic	8

Logic Unit; Design of Accumulator Control Logic Design: Control Organization; Hard-Wired Control; Micro Program Control; Control Of Processor Unit; PLA Control.	
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Course Outcome:

At the end of course, the student will be able to

Course Outcome(CO)		Bloom's Level
CO1	Acquire knowledge about different types of diodes and circuits.	K2
CO2	Apply the knowledge of gates, Boolean algebra and operational amplifier in designing logical and integrated circuits.	K3
CO3	Analyze the logical, integrated, and operational amplifier-based circuits.	K4
CO4	Evaluate the different circuits and compare their performance.	K5
CO5	Design ALU and control unit.	K6

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create

Text Books and Reference Books:

- Schilling Donald L. and Belove E., "Electronics Circuits- Discrete and Integrated", 3rd Ed., McGraw Hill, 1989, Reprint 2008.
- Millman Jacob, Halkias Christos C. and Parikh C., "Integrated Electronics", 2nd Ed., McGraw-Hill, 2009.
- Taub H. and Mothibi Suryaprakash, Millman J., "Pulse, Digital and Switching Waveforms", 2nd Ed., McGraw-Hill, 2007.
- Mano Morris, "Digital Logic and Computer Design", 5th Ed., Pearson Education, 2005.
- Lee Samual, "Digital Circuits and Logic Design", 1st Ed., PHI, 1998.
- Malvin Albert & David J. Bates, "Electronic Principles", 7th edition, Tata McGraw Hill, 2007.
- De Debashis, "Basic of Electronics", 1st Ed., Pearson Education, 2008
- Floyd and Jain, "Digital Fundamentals", Pearson Education, 2006.
- <https://onlinecourses.nptel.ac.in/>

CCS 101: INTRODUCTION TO COMPUTER SCIENCE

Course Objectives: The objectives of the course are as under:

1. To learn about computers and its basic architecture
2. To learn to use of number systems
3. To understand the use of pseudocodes and flowcharts
4. To study about different softwares and operating systems
5. To learn the basics of security and internet.

Syllabus

Content	Contact Hours
UNIT I:	8
Introduction To Computer And Its Architecture: Introduction to computer and its characteristics, Computer Architecture, Generations, Classifications, Applications, Central Processing Unit and Memory, Communication between various Units, Processor Speed, Multiprocessor System, Peripheral Buses, Motherboard Demonstration. Number Systems: Introduction and type of Number System, Conversion between Number System, Arithmetic Operations in different Number System, Signed and Unsigned Number System.	
UNIT II	8
Computational Problem Solving: Program Development Cycle, Pseudocode, Flowchart, Representing Information as Bits, Binary System, Storing Integers, Storing Fractions, Examples of Computational Problems, Iterative and Recursive Approaches to Solve Computational Problems, Easy and Hard Computational Problems	
UNIT III	8
Memory And Various Input And Output Devices: Introduction to Memory, Input and Output Devices, Memory Hierarchy, Primary Memory and its Types, Secondary Memory, Classification of Secondary Memory, Various Secondary Storage Devices and their Functioning. Introduction To System Softwares And Programming Languages: Classification of Computer Languages, Introduction of Operating System, Evolution, Type and Function of OS, Unix Commands, Evolution and Classification of programming Language, Feature and Selection of good Programming Language, Development of Program, Algorithm and Flowchart, Program Testing and Debugging, Program Documentation and Paradigms, Characteristics of good Program.	
UNIT IV	8
Windows Operating System And Its Environment: Introduction to GUI based OS, Configuration, Setup, Services, Network Configuration. Linux Operating System And Its Environment: Introduction to Linux OS, Configuration, Setup, Commands – Navigating File System, File Permissions (R/W/X), Access control and super user (sudo) privileges, Scripting basics,	

Bash Shell and Scripting, Network Configuration.	
UNIT V	8
Data Communication, Computer Network And Internet Basics: Data Communication and Transmission media, Multiplexing and Switching, Computer Network and Network Topology, Communication Protocols and Network Devices, Evolution and Basic Internet Term, Getting Connected to Internet and Internet Application, Email and its working, Searching the Web, Languages of Internet, Internet and Viruses. SYSTEM AND NETWORK SECURITY BASICS: Security Services, Security Attacks, and Security Mechanisms, Authentication, Password Strengths and Entropy, Access Control Mechanisms, Read/Write/Execute Permissions and Super User/Administrator Privileges, Introduction of HTTPS and Digital Certificates	

Course Outcomes:

Upon completion of the course, the student will be able to:

	Course Outcomes	Bloom's Level
CO-1	Acquire knowledge about computers and computational problem solving.	K2
CO-2	Design the solutions of computational problems using iterative and recursive methods using flowcharts and pseudo-codes.	K3
CO-3	Solve computational problems in different number systems.	K4
CO-4	Analyse the importance of different types of memory and evaluate the impact of different algorithms on memory.	K3
CO-5	Experiment with different operating systems such as Windows and Linux and write scripts to automate repetitive tasks.	K3

TEXT BOOKS AND REFERENCE BOOKS:

1. "Introduction to Computer Science", Fourth Impression, Pearson Education, ITL Education Solutions Limited, 2009.
2. Nell Dale and John Lewis, "Computer Science Illuminated", Jones and Bartlett Publishers.
3. Robert Sedgewick and Kevin Wayne, "Computer Science", Addison-Wesley.
4. <https://onlinecourses.nptel.ac.in/>

CAI 101: INTRODUCTION TO COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Course Objectives: The objectives of this course is

1. To learn about computers and its basic architecture
2. To learn to use of number systems
3. To understand the use of pseudocodes and flowcharts
4. To learn about artificial intelligence (AI) and study its interesting features
5. To know how to use AI in modern day applications

Syllabus

Content	Contact Hours
UNIT I:	8
Introduction To Computer And Its Architecture: Introduction to computer and its characteristics, Computer Architecture, Generations, Classifications, Applications, Central Processing Unit and Memory, Communication between various Units, Processor Speed, Multiprocessor System, Peripheral Buses, Motherboard Demonstration. Number Systems: Introduction and type of Number System, Conversion between Number System, Arithmetic Operations in different Number System, Signed and Unsigned Number System.	
UNIT II	8
Computational Problem Solving: Program Development Cycle, Pseudocode, Flowchart, Representing Information as Bits, Binary System, Storing Integers, Storing Fractions, Examples of Computational Problems, Iterative and Recursive Approaches to Solve Computational Problems, Easy and Hard Computational Problems	
UNIT III	8
Memory And Various Input And Output Devices: Introduction to Memory, Input and Output Devices, Memory Hierarchy, Primary Memory and its Types, Secondary Memory, Classification of Secondary Memory, Various Secondary Storage Devices and their Functioning. Introduction To System Softwares And Programming Languages: Classification of Computer Languages, Introduction of Operating System, Evolution, Type and Function of OS, Unix Commands, Evolution and Classification of programming Language, Feature and Selection of good Programming Language, Development of Program, Algorithm and Flowchart, Program Testing and Debugging, Program Documentation and Paradigms, Characteristics of good Program.	
UNIT IV	8
Introduction to Artificial Intelligence (AI): definition, foundation and history of AI, types of AI, intelligent agents, structure of intelligent agents, introduction to soft computing, introduction and operations on fuzzy sets, nature inspired computing and algorithms.	

UNIT V	8
AI terminologies and basic concepts, searching for solutions, search strategies. Applications and future of AI, ethical issues, impact of AI on public life: understanding applications of AI in healthcare, gaming, finance, data security, social media, travel & transport, automotive industry, robotics, AI in entertainment, agriculture, E-commerce and education	

Course Outcomes:

Upon completion of the course, the student will be able to:

	Course Outcomes	Bloom's Level
CO-1	Acquire knowledge about computers and computational problem solving.	K2
CO-2	Design the solutions of computational problems using iterative and recursive methods using flowcharts and pseudo-codes.	K3
CO-3	Solve computational problems in different number systems.	K4
CO-4	Understand the concept of AI.	K2
CO-5	Understand the applications of AI in different sectors	K2

TEXT BOOKS AND REFERENCE BOOKS:

1. "Introduction to Computer Science", Fourth Impression, Pearson Education, ITL Education Solutions Limited, 2009.
2. Nell Dale and John Lewis, "Computer Science Illuminated", Jones and Bartlett Publishers.
3. Robert Sedgewick and Kevin Wayne, "Computer Science", Addison-Wesley.
4. Stuart Russell and Peter Norvig. Artificial Intelligence: A modern approach, Pearson Education, Inc. 2010.
5. Elaine Rich and Kevin Knight. "Artificial Intelligence". Tata McGraw-Hill, 2003
6. F.O. Karry, C.D. Silva, Soft Computing and Intelligent Systems Design, Pearson, 2009.
7. <https://onlinecourses.nptel.ac.in/>

CCS 203: DATA STRUCTURES

Course Objectives: In this course on data structures, the student will be introduced to linked lists, stacks, queues, etc. to understand how the data is managed in the memory. Additionally, studies on searching, sorting, trees and graphs will be introduced.

Syllabus

Content	Contact Hours
UNIT I:	8
Introduction: Basic Terminology, Elementary Data Organization, Built in Data Types in C. Algorithm, Efficiency of an Algorithm, Time and Space Complexity, Asymptotic notations: Big Oh, Big Theta and Big Omega, Time-Space trade-off. Abstract Data Types (ADT) Arrays: Definition, Single and Multidimensional Arrays, Representation of Arrays: Row Major Order, and Column Major Order, Derivation of Index Formulae for 1-D,2-D,3-D and n-D Array Application of arrays, Sparse Matrices and their representations. Linked lists: Array Implementation and Pointer Implementation of Singly Linked Lists, Doubly Linked List, Circularly Linked List, Operations on a Linked List. Insertion, Deletion, Traversal, Polynomial Representation and Addition Subtraction & Multiplications of Single variable & Two variables Polynomial.	
UNIT II	8
Stacks: Abstract Data Type, Primitive Stack operations: Push & Pop, Array and Linked Implementation of Stack in C, Application of stack: Prefix and Postfix Expressions, Evaluation of postfix expression, Iteration and Recursion-Principles of recursion, Tail recursion, Removal of recursion Problem solving using iteration and recursion with examples such as binary search, Fibonacci numbers, and Hanoi towers. Tradeoffs between iteration and recursion. Queues: Operations on Queue: Create, Add, Delete, Full and Empty, Circular queues, Array and linked implementation of queues in C, Dequeue and Priority Queue.	
UNIT III	8
Searching: Concept of Searching, Sequential search, Index Sequential Search, Binary Search. Concept of Hashing & Collision resolution Techniques used in Hashing. Sorting: Insertion Sort, Selection, Bubble Sort, Quick Sort, Merge Sort, Heap Sort and Radix Sort.	
UNIT IV	8
Trees: Basic terminology used with Tree, Binary Trees, Binary Tree Representation: Array Representation and Pointer (Linked List) Representation, Binary Search Tree, Strictly Binary Tree, Complete Binary Tree. A Extended Binary Trees, Tree Traversal algorithms: Inorder, Preorder and Postorder, Constructing Binary Tree from given Tree Traversal, Operation of Insertion, Deletion, Searching & Modification of data in Binary Search . Threaded Binary	

trees, Traversing Threaded Binary trees. Huffman coding using Binary Tree. Concept & Basic Operations for AVL Tree, B Tree & Binary Heaps	
UNIT V	8
Graphs: Terminology used with Graph, Data Structure for Graph Representations: Adjacency Matrices, Adjacency List, Adjacency. Graph Traversal: Depth First Search and Breadth First Search, Connected Component, Spanning Trees, Minimum Cost Spanning Trees: Prims and Kruskal algorithm. Transitive Closure and Shortest Path algorithm: Warshal Algorithm and Dijkstra Algorithm.	

Course Outcomes:

Upon completion of the course, the student will be able to:

	Course Outcomes	Bloom's Level
CO-1	Describe how arrays, linked lists, stacks, queues, trees, and graphs are represented in memory, used by the algorithms and their common applications.	K1, K2
CO-2	Discuss the computational efficiency of the sorting and searching algorithms.	K2
CO-3	Implementation of Trees and Graphs and perform various operations on these data structure.	K3
CO-4	Understanding the concept of recursion, application of recursion and its implementation and removal of recursion.	K4
CO-5	Identify the alternative implementations of data structures with respect to its performance to solve a real world problem.	K5, K6

TEXT BOOKS AND REFERENCE BOOKS:

1. Aaron M. Tenenbaum, Yedidyah Langsam and Moshe J. Augenstein, "Data Structures Using C and C++", PHI Learning Private Limited, Delhi India.
2. Gilberg ,Forouzan, Data Structures: A Pseudocode Approach with C 3rd edition , Cengage Learning publication.
3. Horowitz and Sahani, "Fundamentals of Data Structures", Galgotia Publications Pvt Ltd Delhi India.
4. Lipschutz, "Data Structures" Schaum's Outline Series, Tata McGraw-hill Education (India) Pvt. Ltd.
5. Thareja, "Data Structure Using C" Oxford Higher Education.
6. AK Sharma, "Data Structure Using C", Pearson Education India.
7. Rajesh K. Shukla, "Data Structure Using C and C++" Wiley Dreamtech Publication.
8. Michael T. Goodrich, Roberto Tamassia, David M. Mount "Data Structures and Algorithms in C++", Wiley India.
9. P. S. Deshpandey, "C and Data structure", Wiley Dreamtech Publication.
10. R. Kruse etal, "Data Structures and Program Design in C", Pearson Education.
11. Berztiss, AT: Data structures, Theory and Practice, Academic Press.
12. Jean Paul Trembley and Paul G. Sorenson, "An Introduction to Data Structures with applications", McGraw Hill.
13. Adam Drozdek "Data Structures and Algorithm in Java", Cengage Learning

CCS 204: WEB PROGRAMMING AND PYTHON

Course Objectives: The objective of the course is to understand and implement Python for different internet applications.

Syllabus

Content	Contact Hours
UNIT I:	8
Introduction: Basics of Internet, World Wide Web, HTTP Protocol, Universal Resource Locator, Web Server, Different Types of Web Servers, Domain Name Server, Web Server Configuration, Internet Browser, Web Document and Mark-Up Language, Hypertext Mark-Up Language, Hypermedia, Web Site Organization, Content Organization, Web Server on Different Operating System Platforms, Web Applications, Web Interface, Web Standards & Accessible Design. Static Web Pages: Web Page, Static Web Page, Hypertext Mark-Up Tags, Handling Font Style, Types, Size, Colour Etc., Handling Table, List, Images, Graphics, Menu Etc; Forms, Input Text Box, Drop Down Menu, Name Variable, Cookie Management, Session Management, Animation, Structure Web Pages, Image Mapping, Link Setup In Image, Frames, Structuring Web Pages Using Frames, Multimedia Handling, Linking to Pages	
UNIT II	8
Dynamic Web Pages and Scripting: Scripting Language, Dynamic Pages and Forms Validation, Validation of Input Text Box, Dynamic Drop Down Menu, Validation and Accessing Name Variable-Value Pair, Cookie Management Through Scripting, Session Management through Scripting, Animation through Scripting, Dynamic Image Mapping Through Scripting, Link Handling through Scripting, Multimedia Handling through Scripting;	
UNIT III	8
Style Sheet and Web Publishing: Web Page Designing using Style Sheet, Different Types of Style Sheet, Defining Different Styles, Export and Importing Style Sheet, Cascade Style Sheet. Web Hosting and Publishing - Different Steps of Web Hosting and Publishing, Documents Interchange Standards, Website Evaluation, Components of Web Publishing, Document Management, Search Engines, and Registration of a Web Site on Search Engines, Publishing Tools. Introduction to Python: Python compiler and PVM, Python installation and environment, Datatypes in Python, Operators in Python, Input/Output, Control Statements Functions in Python: Built in and User defined functions, Writing your own functions, Importing functions, Modules and Packages	
UNIT IV	8
Python Program Flow Control Conditional blocks: if, else and else if, Simple for loops in python, For loop using ranges, string, list and dictionaries. Use of while loops in python, Loop manipulation using pass, continue, break and else. Programming using Python conditional and loop blocks. Python Complex data types: Using string data type and string operations, Defining list and list slicing, Use of Tuple data type. String, List and Dictionary, Manipulations Building blocks of python programs, string manipulation	

methods, List manipulation. Dictionary manipulation, Programming using string, list and dictionary in-built functions. Python Functions, Organizing python codes using functions.	
UNIT V	8
Python File Operations: Reading files, Writing files in python, Understanding read functions, read (), readline (), readlines (). Understanding write functions, write () and writelines () Manipulating file pointer using seek Programming, using file operations. Python packages: Simple programs using the built-in functions of packages matplotlib, numpy, pandas etc. GUI Programming: Tkinter introduction, Tkinter and PythonProgramming, Tk Widgets, Tkinter examples. Python programming with IDE.	

Course Outcomes:

Upon completion of the course, the student will be able to:

	Course Outcomes	Bloom's Level
CO-1	Acquire knowledge about the basics of web pages, need of web server, configuration, client and server side scripting, style of web pages and script programming.	K1, K2
CO-2	Install and configure the web server and apply the knowledge of programming to develop web application pages using html, style sheets, client and server side scripts using script programming.	K1, K2
CO-3	Analyse given problem for the requirement of html, style sheets, client side or server side script with different programming constructs. Interpret the fundamental Python syntax and semantics and be fluent in the use of Python control flow statements.	K3
CO-4	Understand and apply Python program flow controls	K2, K3
CO-5	Understand and apply Python file operations and Python Packages	K2, K3

TEXT BOOKS AND REFERENCE BOOKS:

1. Wesley J. Chun, "Core Python Applications Programming", 3rd Edition , Pearson Education, 2016
2. Lambert, Fundamentals of Python: First Programs with MindTap, 2nd 1st edition , Cengage Learning publication
3. Charles Dierbach, "Introduction to Computer Science using Python", Wiley, 2015
4. Jeeva Jose &P.SojanLal, "Introduction to Computing and Problem Solving with PYTHON", Khanna Publishers, New Delhi, 2016
5. Downey, A. et al., "How to think like a Computer Scientist: Learning with Python", John Wiley, 2015
6. Mark Lutz, "Learning Python", 5th edition, Orelly Publication, 2013, ISBN 978- 1449355739
7. John Zelle, "Python Programming: An Introduction to Computer Science", Second edition, Course Technology Cengage Learning Publications, 2013, ISBN 978- 1590282410
8. Michel Dawson, "Python Programming for Absolute Beginners" , Third Edition, Course Technology Cengage Learning Publications, 2013, ISBN 978-1435455009

9. David Beazley, Brian Jones., “Python Cookbook”, Third Edition, Orelly Publication, 2013, ISBN 978-1449340377
10. <https://onlinecourses.nptel.ac.in/>

CAS 205: ENGLISH FOR TECHNICAL WRITING

The objectives of this course are to enable students to:

1. To provide learning environment to practice listening, speaking, reading and writing skills.
2. To assist the students to carry on the tasks and activities through guided instructions and materials.
3. To effectively integrate English language learning with employability skills and training.
4. To provide hands-on experience through case-studies, mini-projects, group and individual presentations.

Content	Contact Hours
UNIT I: Vocabulary Building	8
The concept of Word Formation. Root words from foreign languages and their use in English. Acquaintance with prefixes and suffixes from foreign languages in English to form derivatives. Synonyms, antonyms, and standard abbreviations.	
UNIT II: Basic Writing Skills	8
Sentence Structures. Use of phrases and clauses in sentences. Importance of proper punctuation. Creating coherence. Organizing principles of paragraphs in documents. Techniques for writing precisely.	
UNIT III: Identifying Common Errors in Writing	8
Subject-verb agreement. Noun-pronoun agreement. Misplaced modifiers. Articles. Prepositions. Redundancies. Clichés	
UNIT IV: Nature and Style of Sensible Writing	8
Describing. Defining. Classifying. Providing examples or evidence. Writing introduction and conclusion.	
UNIT V: Writing Practices	8
Comprehension, Precis Writing, Essay Writing	

TEXTBOOK AND REFERENCES:

1. Communication Skills in English. Anjana Tiwari. Khanna Book Publishing. 2023.
2. Practical English Usage. Michael Swan. OUP. 1995.
3. Remedial English Grammar. F.T. Wood. Macmillan. 2007
4. On Writing Well. William Zinsser. Harper Resource Book. 2001
5. Study Writing. Liz Hamp-Lyons and Ben Heasley. Cambridge University Press. 2006.
6. Communication Skills. Sanjay Kumar and PushpLata. Oxford University Press. 2011.
7. Exercises in Spoken English. Parts. I-III. CIEFL, Hyderabad. Oxford University Press
8. <https://onlinecourses.nptel.ac.in/>

Course Outcomes:

Upon completion of the course, the student will be able to:

	Course Outcomes
CO-1	Understand vocabulary building techniques
CO-2	Understand basic writing skills

CO-3	Apply the writing skills to identify common errors in writing
CO-4	Understand and practice the importance of writing a good introduction and conclusion
CO-5	Apply the writing skills to writing comprehension, précis and essay writing

CCS 251: COMPUTER ORGANIZATION LAB
List of Practicals (Indicative & not limited to)

1. Study of BJT Characteristics
2. Study of CE Amplifier
3. Study of RC Coupled / Tuned Amplifier
4. Study of FET Characteristics
5. Study of Diode Clipper Circuits
6. Study of Diode Clamper Circuits
7. Study and Implement RC Low Pass and High Pass Filter Circuits
8. Study and Implement RC Integrator Circuits
9. Study and Implement RC Differentiator Circuits
10. Full and Half-Adder/ Half-subtractor Circuits using a serial Input
11. 4-Bit Gray to Binary/ Binary to Gray Code convertor using Select input
12. Logic expression with the Help of MUX IC 74153
13. Flip-flops using NAND/ NOR Gate
14. Modulo-7 Ripple Counter
15. 4-Bit Shift Left/Right Register
16. Sequence Generator

CCS 254: WEB PROGRAMMING AND PYTHON LAB
Suggested list of experiments (Indicative & not limited to)

1. To prepare the web page using hypertext markup language
2. To study and setup the web server for implementation
3. To learn client side scripting
4. To learn server side scripting
5. To apply style to the web pages
6. To implement functions for files
7. To implement dictionary
8. Python programming

CEE 152: ELECTRICAL AND ELECTRONICS ENGINEERING LAB

List of Experiments (indicative and not limited to)

1. To study the V-I characteristics of an incandescent lamp.
2. To measure single phase power by using three ammeter method.
3. To measure the single-phase power by using three voltmeter method.
4. To perform short circuit test on a single-phase transformer.
5. To perform open circuit test on a single-phase transformer.
6. To measure three phase power by using two wattmeter method.
7. To study the PN Junction diode characteristics.
8. To design the half wave and full wave rectifiers circuits.
9. To study CB, CE, CC input and output characteristics.

CCS 152: COMPUTATIONAL THINKING THROUGH PROGRAMMING LAB

Programming for problem solving lab includes coding, experiments in C/C++/MATLAB (indicative and not limited to)

1. WAP that accepts the marks of 5 subjects and finds the sum and percentage marks obtained by the student.
2. WAP that calculates the Simple Interest and Compound Interest. The Principal, Amount, Rate of Interest and Time are entered through the keyboard.
3. WAP to calculate the area and circumference of a circle.
4. WAP that accepts the temperature in Centigrade and converts into Fahrenheit using the formula $C/5 = (F-32)/9$.
5. WAP that swaps values of two variables using a third variable.
6. WAP that checks whether the two numbers entered by the user are equal or not.
7. WAP to find the greatest of three numbers.
8. WAP that finds whether a given number is even or odd.
9. WAP that tells whether a given year is a leap year or not.
10. WAP that accepts marks of five subjects and finds percentage and prints grades according to the following criteria:
 - Between 90-100% ---- Print 'A'
 - 80-90% ----- Print 'B'
 - 60-80% ----- Print 'C'
 - Below 60%-----Print 'D'
11. WAP that takes two operands and one operator from the user, perform the operation, and prints the result by using Switch statement.
12. WAP to print the sum of all numbers up to a given number.
13. WAP to find the factorial of a given number.
14. WAP to print sum of even and odd numbers from 1 to N numbers.
15. WAP to print the Fibonacci series.
16. WAP to check whether the entered number is prime or not.
17. WAP to find the sum of digits of the entered number.
18. WAP to find the reverse of a number.
19. WAP to print Armstrong numbers from 1 to 100.
20. WAP to convert binary number into decimal number and vice versa.
21. WAP that simply takes elements of the array from the user and finds the sum of these elements.
22. WAP that inputs two arrays and saves sum of corresponding elements of these arrays in a third array and prints them.
23. WAP to find the minimum and maximum element of the array.
24. WAP to search an element in an array using Linear Search.
25. WAP to sort the elements of the array in ascending order using Bubble Sort technique.
26. WAP to add and multiply two matrices of order nxn.
27. WAP that finds the sum of diagonal elements of a mxn matrix.
28. WAP to implement strlen (), strcat (), strcpy () using the concept of Functions.
29. Define a structure data type TRAIN_INFO. The type contain Train No.:

integer type Train name: string Departure Time: aggregate type TIME
 Arrival Time: aggregate type TIME Start station: string End station: string
 The structure type Time contains two integer members: hour and minute.
 Maintain a train timetable and implement the following operations:

- a. List all the trains (sorted according to train number) that depart from a particular section.
 - b. List all the trains that depart from a particular station at a particular time.
 - c. List all the trains that depart from a particular station within the next one hour of a given time.
 - d. List all the trains between a pair of start station and end station.
30. WAP to swap two elements using the concept of pointers.
 31. WAP to compare the contents of two files and determine whether they are same or not.
 32. WAP to check whether a given word exists in a file or not. If yes then find the number of times it occurs.

Note:

- a) *The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner*
- b) *The subject teachers are suggested to use the concept of project based learning. The subject teacher may give certain use cases/case studies where student is able to apply multiple concepts in one single program*

Mapping with Virtual Lab

Name of the Lab	Name of the Experiment
Problem Solving Lab	Numerical Representation
	Beauty of Numbers
	More on Numbers
	Factorials
	String Operations
	Recursion
	Advanced Arithmetic
	Searching and Sorting
	Permutation
	Sequences

Course Outcomes:

At the end of course, the student will be able to:

Course Outcome		Bloom's Level
CO 1	Able to implement the algorithms and draw flowcharts for solving Mathematical and Engineering problems.	K3, K4
CO 2	Demonstrate an understanding of computer programming language concepts.	K3, K2
CO 3	Ability to design and develop Computer programs, analyzes, and interprets the concept of pointers, declarations, initialization, operations on pointers and their usage.	K6, K4

CO 4	Able to define data types and use them in simple data processing applications he/she must be able to use the concept of array of structures.	K1, K5
CO 5	Develop confidence for self-education and ability for life-long learning needed for Computer language.	K3, K4

K1- Remember, K2- Understand, K3- Apply, K4- Analyze , K5- Evaluate , K6- Create

CCS 253: DATA STRUCTURE LAB

List of Experiments (Indicative & not limited to)

- 1. Implementing Sorting Techniques:** Bubble Sort, Insertion Sort, Selection Sort, Shell Sort, Radix Sort, Quick sort
- 2. Implementing Searching and Hashing Techniques:** Linear search, Binary search, Methods for Hashing: Modulo Division, Digit Extraction, Fold shift, Fold Boundary, Linear Probe for Collision Resolution. Direct and Subtraction hashing
- 3. Implementing Stacks:** Array implementation, Linked List implementation, Evaluation of postfix expression and balancing of parenthesis, Conversion of infix notation to postfix notation
- 4. Implementing Queue:** Linked List implementation of ordinary queue, Array implementation of circular queue, Linked List implementation of priority queue, Double ended queue
- 5. Implementing Linked List:** Singly Linked Lists, Circular Linked List, Doubly Linked Lists : Insert, Display, Delete, Search, Count, Reverse(SLL), Polynomial Addition, Comparative study of arrays and linked list
- 6. Implementing Trees:** Binary search tree: Create, Recursive traversal: preorder, post order, in order, Search Largest Node, Smallest Node, Count number of nodes, Heap: Min Heap, Max Heap: reheap Up, reheap Down, Delete, Expression Tree, Heapsort
- 7. Implementing Graphs:** Represent a graph using the Adjacency Matrix, BFS, Find the minimum spanning tree (using any method Kruskal's Algorithm or Prim's Algorithm) Self Learning Topics : Shortest Path Algorithm

CAS 253: ENGLISH LANGUAGE LAB

Course Objectives:

1. To facilitate software based learning to provide the required English Language proficiency to students.
2. To acquaint students with specific dimensions of communication skills i.e. Reading, Writing, Listening, Thinking and Speaking.
3. To train students to use the correct and error-free writing by being well versed in rules of English grammar.
4. To cultivate relevant technical style of communication and presentation at their work place and also for academic uses.
5. To enable students to apply it for practical and oral presentation purposes by being honed up in presentation skills and voice-dynamics.

Professional Communication Lab shall have two parts:

1- Interactive Communication Skills:

Students should practice the language with variety of activities and exercises based on employability skills. Interactive and communicative practical with emphasis on oral presentation/spoken communication, based on International Phonetic Alphabets (I.P.A.)

LIST OF PRACTICALS

1. Group Discussion: Practical based on Accurate and Current Grammatical Patterns.
2. Conversational Skills for Interviews under suitable Professional Communication Lab conditions with emphasis on Kinesics.
3. Communication Skills for Seminars/Conferences/Workshops with emphasis on Paralinguistic/Kinesics.
4. Presentation Skills for Technical Paper/Project Reports/proposals based on proper Stress and Intonation Mechanics
5. Official/Public Speaking practice sessions based on suitable Rhythmic Patterns.
6. Theme Presentation/Keynote Presentation based on correct methodologies of argumentation
7. Individual Speech Delivery/Conferencing with skills to defend Interjections/Quizzes.
8. Argumentative Skills/Role Play Presentation with Stress and Intonation.
9. Comprehension Skills based on Reading and Listening Practical's on a model Audio
10. Startup presentations, Video portfolio, Extempore, Roleplay, Just A Minute (JAM) etc.

2- Computer assisted software based Language Learning:

Software based self guided learning to provide the required English language proficiency to students from an employability and career readiness stand point. The software should align to Common European Framework of Reference for Languages (CEFR) and deliver a CEFR level–B2 upon completion

Course Outcomes:

Upon completion of the course, the student will be able to:

	Course Outcomes
CO-1	Students will be able to understand the basic objective of the course by being acquainted with specific dimensions of communication skills i.e. Reading, Writing, Listening, Thinking and Speaking.
CO-2	Students would be able to create substantial base by the formation of strong professional vocabulary for its application at different platforms and through numerous modes as Comprehension, reading, writing and speaking etc.
CO-3	Students will apply it at their workplace for writing purposes such as Presentation/official drafting/administrative communication and use it for document/project/report/research paper writing.
CO-4	Students will be made to evaluate the correct and error-free writing by being well-versed in rules of English grammar and cultivate relevant technical style of communication & presentation at their workplace and also for academic uses.
CO-5	Students will apply it for practical and oral presentation purposes by being honed up in presentation skills and voice dynamics. They will apply techniques for developing interpersonal communication skills and positive attitude leading to their professional competence.

Suggested Softwares:

- *Oxford Achiever* by Oxford University Press.
- *Cambridge English Empower* by Cambridge University Press.
- *MePro*. By Pearson India Education Services Pvt .Ltd.
- *New Interactions* by McGraw-Hill India.

Reference Books:

1. Word Power Made Easy by Norman Lewis, W.R.Goyal Pub. & Distributors, 2009, Delhi.
2. Manual of Practical Communication by L.U.B. Pandey; A.I.T.B.S. Publications India Ltd.; KrishanNagar, 2013, Delhi
3. Practical Communication Process & Practice, LU.B. Pandey: A.I.T.B.S. Pub. India Ltd Krishna Nagar, Delhi, 2013.
4. English Grammar and Usage by R.P.Sinha, Oxford University Press, 2005, New Delhi.
5. English Grammar, Composition and Usage by N.K.Agrawal & F.T.Wood, Macmillan India Ltd., NewDelhi.
6. Effective Communication Skill, Kulbhusan Kumar, R S Salaria, Khanna Publishing House
7. English Grammar & Composition by Wren & Martin, S.Chand & Co.Ltd., NewDelhi.
8. Communication Skills for Engineers and Scientists, Sangeeta Sharma et.al. PHILearning Pvt. Ltd, 2011, NewDelhi.
9. Personality Development, Harold R. Wallace & L.Ann Masters, Cengage Learning, New Delhi
10. Personality Development & Soft Skills, Barun K.Mitra, Oxford University Press, 2012 New Delhi.
11. Business Correspondence and Report Writing by Prof. R.C.Sharma & Krishna Mohan, Tata Mc Graw Hill & Co.Ltd., 2001, New Delhi.
12. Developing Communication Skills by Krishna Mohan, Meera Bannerji-Macmillan India

Ltd. 1990, Delhi.

13. SpokenEnglish A manual of Speech and Phonetics by R.K. Bansal & J.B. Harrison, Orient Blackswan, 2013, NewDelhi.
14. Business English by KenTaylor, Orient Blackswan, 2011, NewDelhi
15. <https://onlinecourses.nptel.ac.in/>

CME 152: ENGINEERING VISUALIZATION & DESIGN LAB

The objective of this lab is to learn the basics of engineering drawing and implement the knowledge of this subject as an engineering design. The lab curriculum is designed for 1st year UG students for all engineering specializations. At the end of the course students will be able to implement their ideas in various applications and create job opportunities.

UNIT - I

Lines Lettering and Dimensioning: Types of lines, Lettering, Dimensioning, Geometrical Constructions, Polygons. Scales: Plain scales, Diagonal scales, Scale of chords.

UNIT - II

Curves used in Engineering Practice: Ellipse, Parabola, Hyperbola, normal and tangents to these curves, Involute, Cycloid, Epi-cycloid, Hypocycloid, Spiral, Helix on cone and cylinder.

UNIT- III

Orthographic projection of points: Principles of Orthographic projection, Projections of points. Projections of Lines: Projections of a line parallel to one of the reference planes and inclined to the other, line inclined to both the reference planes, Traces, Projections of Planes: Projections of a plane perpendicular to one of the reference planes and inclined to the other, Oblique planes.

UNIT - IV

Projections of Solids: Projections of solids whose axis is parallel to one of the reference planes and inclined to the other, axis inclined to both the planes. Section of Solids: Sectional planes, Sectional views - Prism, pyramid, cylinder and cone, true shape of the section.

UNIT - V

Isometric views: Isometric axis, Isometric Planes, Isometric View, Isometric projection, Isometric views – simple objects. Assembly drawings of the machine parts.

NOTE: Interpretation of drawings: Introduction of CAD package to construct a simple solid model, using a CAD package to construct solid models and generating orthographic, isometric, sectional views with dimensioning, Assembly of components and generation of corresponding drawings. Animation of machines in CAD.

Major Equipments:

Drawing sheet, Drawing board, Mini drafter, HB/1H/2H Pencils, Eraser, Protractor, scale, etc. or using AutoCAD

List of Experiments

S. No. Description

1. Lettering, Dimensioning,
2. Plan Scale, Diagonal Scale, Scale of Chords
3. Geometrical construction of Engineering Curves
4. Introduction of Projection of Points and Lines
5. Traces of lines
6. Projection of Planes
7. Projections of Regular Solids
8. Sectional Views of Solids

- 9. Isometric views
- 10. Orthographic to isometric

Virtual Labs

1: Basics of Engineering Drawing

<https://eg-nitk.vlabs.ac.in/exp/engineering-drawing/simulation.html>

2: Projection of Points

<https://eg-nitk.vlabs.ac.in/exp/projections-of-points/simulation.html>

3: To draw projections of Line inclined with both VP and HP

<https://eg-nitk.vlabs.ac.in/exp/simple-positions/simulation.html>

4: To draw projections of Line inclined with both VP and HP and trace it

<https://eg-nitk.vlabs.ac.in/exp/projections-lines-inclined/simulation.html>

Text Books:

- 1. Engineering Graphics, N.D. Bhatt and V.M. Panchal, Charotar Publishers.
- 2. Jolhe, D. A., Engineering drawing, Tata McGraw Hill

Reference Books:

- 1. Engineering Drawing, Agarwal, B, McGraw Hill Education, 2015, second edition.
- 2. Shah, M. B. and Rana, B. C., Engineering Drawing, Pearson Education, 2009
- 3. K.V. Natarajan, A text book of Engineering Graphics, Dhanalakshmi Publishers, Chennai, 2006.
- 4. AutoCAD 2007 Bible E. Finkelstein, Wiley Publishing Inc.

CME 151: IDEA LAB WORKSHOP

SUGGESTIVE LIST OF EXPERIMENTS:

S. No.	Mechanical Workshop	Duration
1	Introduction to product design To study different tools used in SolidWorks. 2D and 3D part design in SolidWorks. Virtual experiments on PCB Design, Fabrication and Digital Fabrication of flexible circuit board etc.	3 Hours
2	Introduction to Mechanical workshop material, tools and machines To study layout, safety measures and different engineering materials (mild steel, medium carbon steel, high carbon steel, high speed steel and cast iron etc.) used in workshop. To study and use of different types of tools, equipments, devices & machines used in fitting, sheet metal and welding section. To determine the least count of vernier caliper, vernier height gauge, micrometer (Screw gauge) and take different reading over given metallic pieces using these instruments.	3 Hours
3	Machine shop Demonstration of working, construction and accessories for Lathe machine. Perform operations on Lathe - Facing, Plane Turning, step turning, taper turning, threading, knurling and parting. Virtual machining simulation using CAD/CAM tools involves creating a design model, creating workpiece, choosing manufacturing set up, defining machining sequences, generating toolpath, checking the results, and post-processing etc. Generate machining toolpath for complex 3D Shapes suitable for machining on the CNC minimal machine. Simulate the generated toolpath on CNC simulator and estimate machining time.	3 Hours
4	Fitting shop To study about different hand tools used in fitting shop. Preparation of U or V -Shape Male Female Work piece which contains: Filing, Sawing, Drilling, Grinding.	3 Hours
5	Welding Shop Introduction to BI standards and reading of welding drawings. Practice of making following operations Butt Joint, Lap Joint or T Joint by TIG Welding/MIG Welding/Arc Welding Train students with cutting-edge welding technologies on virtual welding simulators.	3 Hours
6	Moulding and Casting Shop Introduction to Patterns, pattern allowances, ingredients of moulding sand and melting furnaces.	3 Hours

- To study the different tools used in Foundry shop.
Demo of mould preparation and Aluminium casting Practice – Study and Preparation of Plastic mould
Virtual experiments on Molding and Casting of parts and components
- 7 **CNC Shop**
Study of main features and working parts of CNC machine and accessories that can be used. 3 Hours
Perform different operations on metal components using any CNC machines
- 8 **Sheet metal shop**
To study different types of Hand tools used in Sheet metal shop. 3 Hours
To prepare a square tray of given dimensions using a Galvanized iron (G.I) sheet.
To prepare a Funnel of given dimensions using a G.I. sheet.
- 9 **To prepare a product using 3D printing**
To study the different types of 3D printing processes. 3 Hours
Design and development of 3D printed part or component on any available 3D printer.
Virtual experiments on different types 3D scanning of objects.
Virtual experiments on different types of 3D Printing processes including their construction, process capabilities, material selection, parameter selection, post processing and defect analysis etc.

The students will be able to		Blooms Taxonomy
CO1	Use various engineering materials, tools, machines and measuring equipments.	K3
CO2	Perform machine operations in lathe and CNC machine.	K3
CO3	Perform manufacturing operations on components in fittings shop.	K3
CO4	Perform operations in welding, moulding, casting and gas cutting.	K3
CO5	Fabricate a job by 3D printing manufacturing technique	K3

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create

Text Books:

1. Basic Workshop Technology: Manufacturing Process, Felix W.; Independently Published, 2019.
2. Workshop Processes, Practices and Materials; Bruce J. Black, Routledge publishers, 5th Edn. 2015.
3. A Course in Workshop Technology Vol I. & II, B.S. Raghuwanshi, Dhanpath Rai & Co., 2015 & 2017.

Reference Books:

1. Workshop Practice by H. S. Bawa, Tata-McGraw Hill, 2004
2. Mechanical Workshop Practice, K C John, PHI
3. Workshop Practice Vol 1, and Vol 2, by Hazra Choudhary, Media promoters and Publications
4. CNC Fundamentals and Programming, By P. M. Agrawal, V. J. Patel, Charotar Publication.

5. Technology of machine tools, Steve F. Krar, Arthur R. Gill and Peter Smid, McGraw Hill Education (India) Pt. Ltd., 2013.
6. Engineering Practices Laboratory Manual, Ramesh Babu.V., VRB Publishers Private Limited, Chennai, Revised edition, 2013 – 2014.
7. Engineering Practices Lab Manual; T.Jeyapoovan, Vikas Pub, 4th Edn.2008.
8. Mechanical Workshop Practice; John K.C., PHI, 2010.
9. Wiring Estimating, Costing and Contracting; Soni P.M. & Upadhyay P.A.; Atul Prakashan, 2021-22.
10. H Gerling, All about Machine Tools, New Age International, 1995.
11. W A J Chapman, Workshop Technology, Oxford IBH, 1975.

Online Resources:

1. Different Trade E-Books (Fitting, Plumbing, Welding, Carpentry, Foundryman, Turner and House Wiring etc.) developed by National Instructional Media Institute, Chennai. Directorate General of Training, Ministry of Skill Development & Entrepreneurship, Govt.of India. (<https://bharatskills.gov.in>).
2. 3D Printing Virtual Simulation Lab (<https://3dp-dei.vlabs.ac.in/Introduction.html>)
3. Fabrication Lab (<https://fab-coep.vlabs.ac.in/Introduction.html>)
4. Mechanisms and Robotics Lab (<https://mr-iitkgp.vlabs.ac.in/Introduction.html>)
5. Micromachining lab (<https://mm-coep.vlabs.ac.in/Introduction.html>)